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Apical shoot elongation of *Lawsonia inermis* L. through tissue culture in plane medium

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Abstract : With the help of standard plant tissue culture technique, elongation of apical shoot of *Lawsonia inermis* L. was carried out in MS media without growth regulations. Small apical shoots of young plants were used as explants.

Keywords : Tissue culture, apical shoot, *Lawsonia inermis* L.

INTRODUCTION

Lawsonia inermis L. belongs to the family Lythracae. It much branched shrub or small tree. It is native of middle east Africa. It is commonly known as Mehandi in Hindi of Egyption Henna. The leaves of the henna plant have a red orange dye molecule Lawsone. Lawsone is a nethaquinon compound. Decoction of its leaf is used as a gargle against sore throat.[1]

MATERIALS & METHOD

In the present investigation apical portion young shoots were used for experiments. The segments of apical shoots were inoculated on Ms medicein without any phytoharmones.

OBSERVATIONS

Apical shoot elongation of *Lawsonia inermis* L. in plane MS medium was observed.

RESULTS & DISCUSSION

In this investigation apical shoots were cultured in plane medium. After 45 days elongation of shoots were obtained. Several attempts have been made to utilize the *in vitro* micropropagation method for rapid clonal multiplication of many tropical, subtropical and temperate fruits[2]. Bud cultured on hormone free basal medium showed elongation, shoot elongation occurred only when cytokinin was omitted from the medicein[3].

Shoot elongation of *Lawsonia inermis* L. in plane Ms medicein (without any growth hormones) in duration of sixth week in observation.

	Apical shoot elongation (in cm)					
MS medium (plane)	1st week	2nd week	3rd week	4th week	5th week	6th week
1	1.5	1.5	1.9	2.0	2.5	3.0
2	1.6	1.6	2.0	2.2	2.7	3.2
3	1.7	1.7	2.1	2.3	2.7	3.4
4	1.8	1.8	2.2	2.4	2.9	3.5
5	1.9	1.9	2.4	2.6	3.0	3.8

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